Palladium
MAKE IT POSSIBLE



ETHIOEMR PLATFORM

Team EMR, Form Studio and AI assistant in one place

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AGENDA

What we built on top of the team's EMR, how it fits together, and what comes next.

01 The problem

02 What the platform is

03 Architecture

04 Form Studio

05 AI assistant and speech

06 What comes next

THE PROBLEM

Four things cost builders and clinicians time today.

1 **Forms are slow to build and check**

Each form is converted from a Bahmni export or a paper template, then checked by hand. A form saves only when every concept it uses exists on the server with the right type.

2 **Clinicians type a lot**

Complaints, findings and form fields are typed during the visit, one field at a time.

3 **Amharic dictation**

Doctors speak Amharic mixed with English terms. General models fail on it: Whisper large-v3 gave 90% character errors on our Amharic test.

4 **Help is scattered**

National guidelines, form guides and field meanings sit in separate PDFs and documents, outside the EMR.

WHAT THE PLATFORM IS

One stack: the team's EMR exactly as on GitHub main, with our modules on top.

9147129

team distro main, unchanged

54 + 2

team apps plus our two modules

1

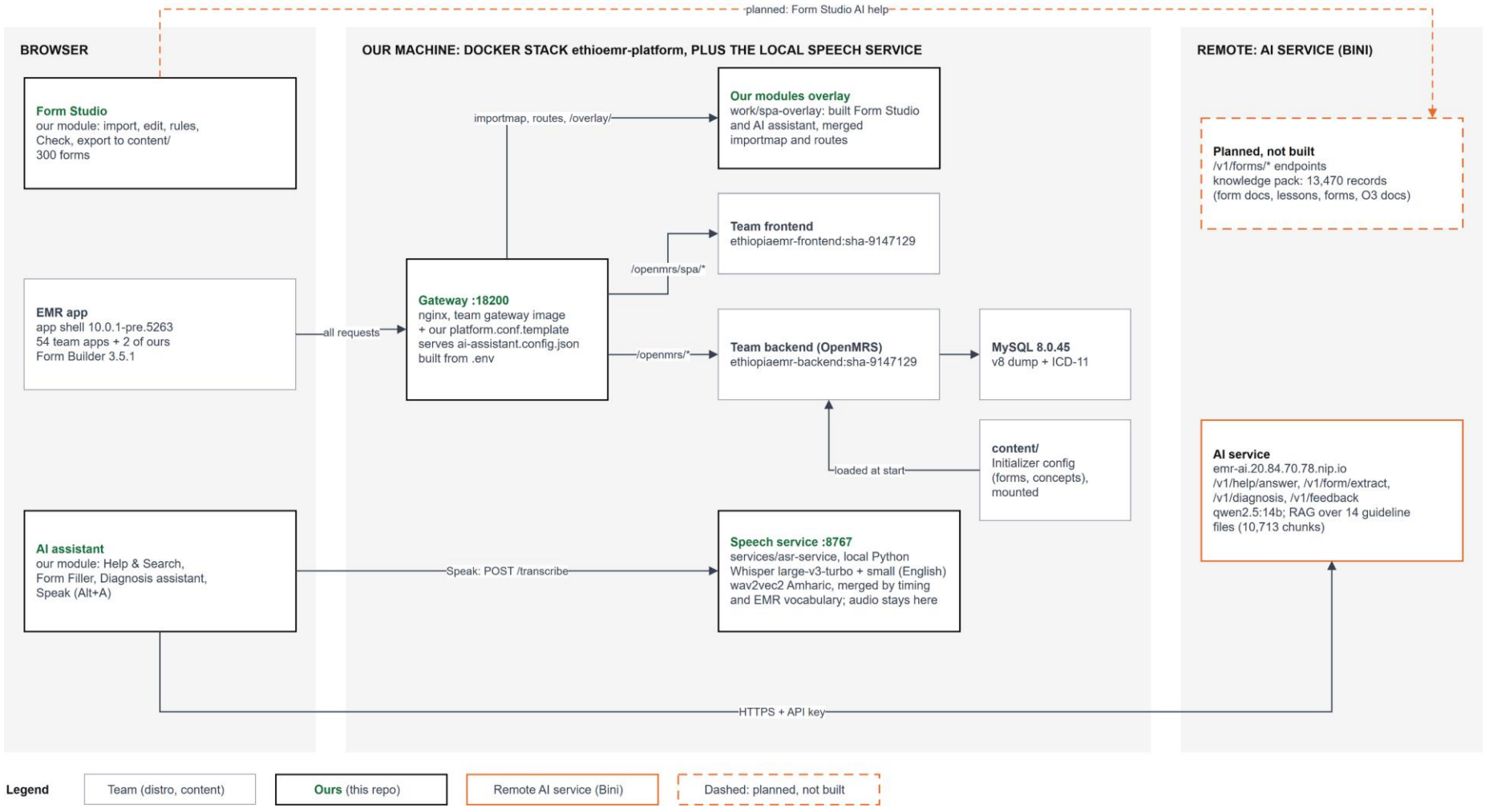
command to start: platform.ps1 up

Part	What it is	Owner
distro/	team images sha-9147129, MySQL 8.0.45	distro team
content/	Initializer config: forms, concepts	content team
gateway/, compose overlay	one entry point on :18200	us
modules/	Form Studio and AI assistant	us
services/asr-service	local speech-to-text, port 8767	us

Open it at **localhost:18200/openmrs/spa**. When our modules are published to npm, each becomes one line in the team's distro and the overlay is no longer needed.

ARCHITECTURE

The browser talks to one gateway. Grey is the team's, black is ours, orange is the remote AI service.



Team images

Frontend and backend images at sha-9147129. MySQL 8.0.45 with the v8 dump and ICD-11.

Our gateway template

Adds our modules and the AI config from .env. The API key is not in git.

Two outside calls

Speech stays on this machine. AI questions go to Bini's service over HTTPS.

FORM STUDIO

An authoring module inside the EMR. It turns a source document into a checked O3 form.

1 Import → **2 Edit** → **3 Rules** → **4 Check** → **5 Export**

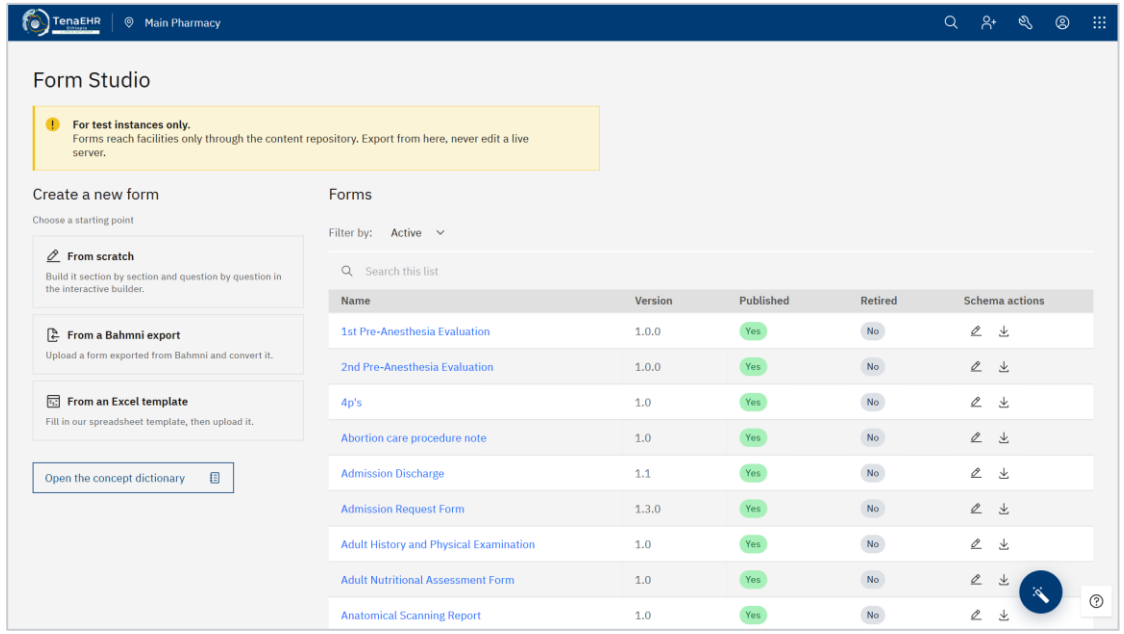
Bahmni export, Excel template or from scratch

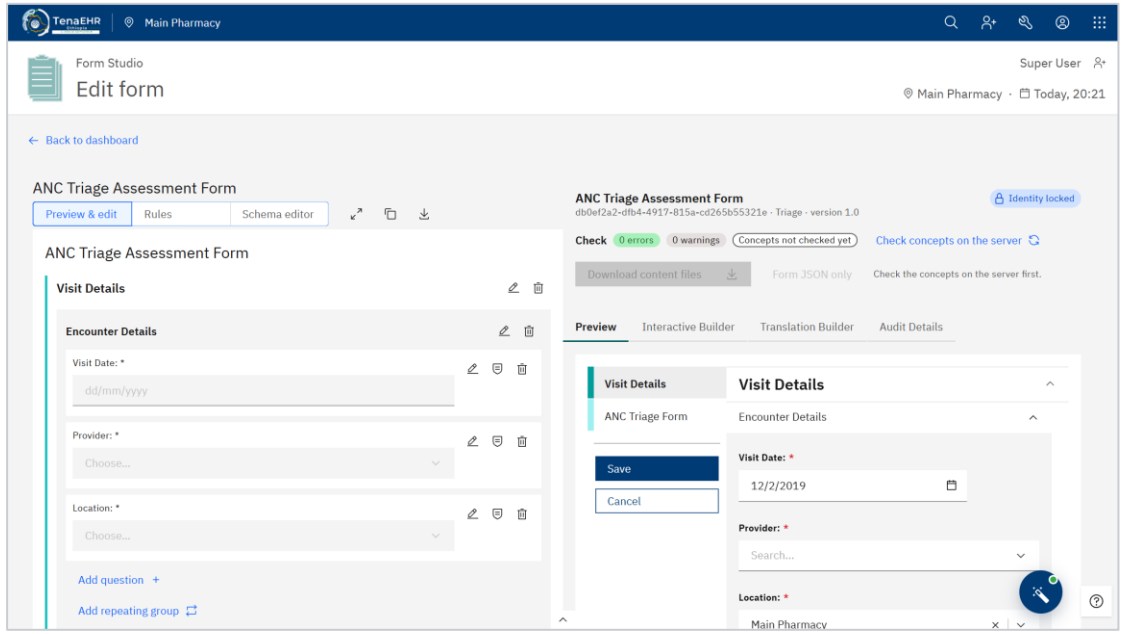
preview and edit, interactive builder

show or hide, warnings, calculated values

errors, warnings, concepts on the server

content files for the content repository





Start page with 300 forms, and the editor with Check showing 0 errors. For test instances only: forms reach facilities through the content repository.

AI ASSISTANT

A panel on every EMR page, opened with Alt+A. Four tools.

1 **Help & Search**

Finds forms, fields and guides. Ask answers from the results and shows its sources.

2 **Form Filler**

Type or speak values; it fills one field at a time or the whole form.

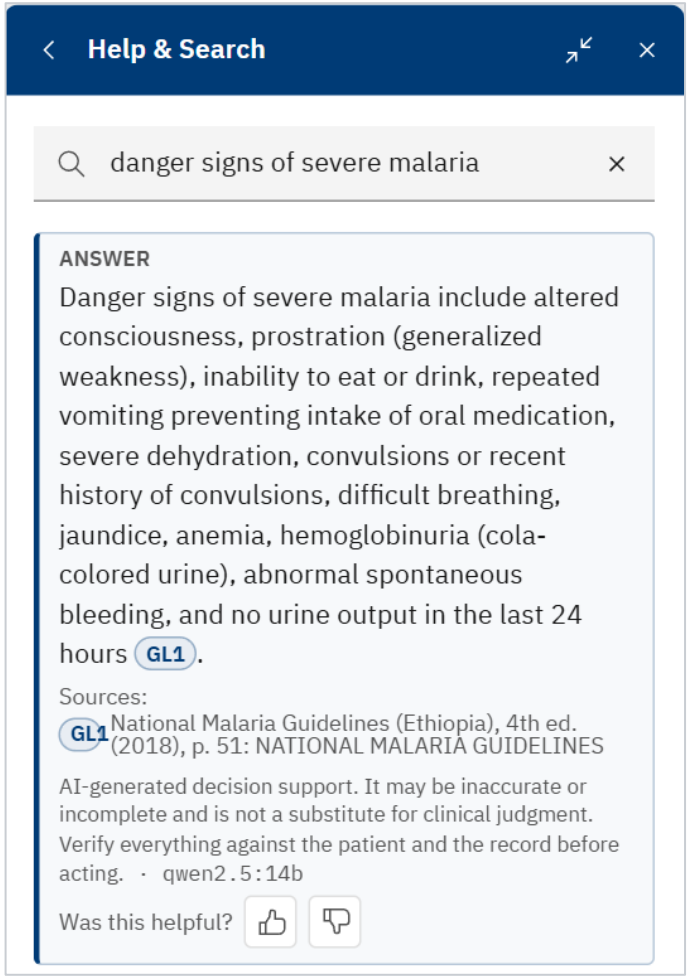
3 **Diagnosis assistant**

A differential from the complaint and this patient's chart, with the records it relied on.

4 **Speak**

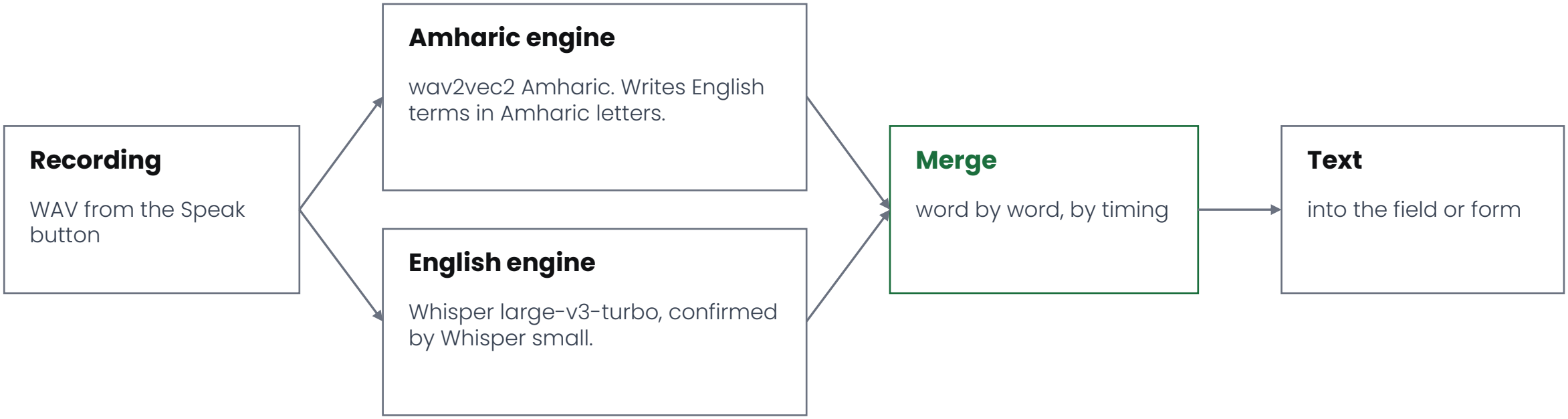
Dictate in Amharic or English. The local speech service writes it out.

RAG: the service retrieves guideline passages, answers only from them and cites them. 14 guideline files, 10,713 passages, model qwen2.5:14b.



SPEECH PIPELINE

Speak runs on our machine. Audio never leaves it, and no LLM touches the text.



An English word replaces the Amharic only if:

1 Whisper is reasonably sure
2 it sounds like what the Amharic engine heard
3 it is in the EMR vocabulary (28k concept and drug words)

Results on our 15 s test: wav2vec2 Amharic 9% character errors; Whisper large-v3 alone 90%. Every output word comes from one of the two engines.

WHAT COMES NEXT

AI help inside Form Studio. Not built yet: it needs Bini's /v1/forms endpoints.

1

Explain errors

When Check finds a problem, the AI explains it in plain words and points to the rule.

2

Draft rules and form JSON

The AI drafts. Our Check verifies each draft before it is accepted.

3

Knowledge from our work

A knowledge pack of 13,470 records: form docs, lessons, our forms, plus O3 docs.

Who does what

Bini: /v1/forms/* endpoints (ask, suggest-concepts, explain-rule, draft-rule, review) and knowledge-pack ingest.

Us: the pack builder, an Ask button in the editor, and Check on anything the user accepts.

Not decided: dates for either side.

Questions & Discussions

Thank you



One stack, team main unchanged, our work on top. Questions and comments are welcome.

Open: localhost:18200/openmrs/spa
Docs: docs/platform/ in the ethioemr-platform repo